Quiz 2 Solutions

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The organizers of a tech conference collected data on the attendees. The companies and ages of the attendees are recorded below:

	20–39	40–59	60+	Total
Apple	41	53	27	**121**
Google	24	67	38	**129**
Meta	34	39	40	**113**
Other	62	71	65	**198**
Total	**161**	**230**	**170**	**561**

An attendee is selected at random from this group. What is the probability that

(a) They are 20–59 and their company is Other?

Here, "and" means an intersection. We need the two cells in the Other row corresponding to ages 20–39 and 40–59.

	20–39	40–59	60+	Total
Apple	41	53	27	121
Google	24	67	38	129
Meta	34	39	40	113
Other	62	71	65	198
Total	161	230	170	561

$$P(\text{20–59} \cap \text{Other}) = \frac{62 + 71}{561} = \boxed{\frac{133}{561}}.$$

(b) Their company is Google or they are 60+?

Let $G = \{\text{Google}\}$ and $S = \{60+\}$. The highlighted cells form the union $G \cup S$. The cell 38 is the intersection $G \cap S$.

	20–39	40–59	60+	Total
Apple	41	53	27	121
Google	24	67	38	129
Meta	34	39	40	113
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Total	161	230	170	561

Method 1: Inclusion–exclusion

$$\begin{aligned} P(G \cup S) &= P(G) + P(S) - P(G \cap S) \\ &= \frac{129}{561} + \frac{170}{561} - \frac{38}{561} \\ &= \boxed{\frac{261}{561}}. \end{aligned}$$

Be careful! Do not simply add $129 + 170$. The 38 attendees who are both Google and 60+ would be counted twice!

Method 2: Count the union directly

$$P(G \cup S) = \frac{24 + 67 + 38 + 27 + 40 + 65}{561} = \boxed{\frac{261}{561}}.$$

(c) Their company is Meta if they are 40–59?

The phrase "if they are 40–59" means that we **condition** on the 40–59 group. Therefore, the denominator is the total number of attendees aged 40–59, which is 230.

	20–39	40–59	60+	Total
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Google	24	67	38	129
Meta	34	39	40	113
Other	62	71	65	198
Total	161	230	170	561

Method 1: Read directly from the table
Among the 230 attendees who are 40–59, 39 are from Meta. Therefore,

$$P(\text{Meta} \mid 40\text{–}59) = \frac{39}{230} = \boxed{\frac{39}{230}}.$$

Method 2: Use the conditional probability formula
Let

$$M = \{\text{the attendee is from Meta}\}, \qquad A = \{\text{the attendee is 40–59}\}.$$

Using the conditional probability formula,

$$P(M \mid A) = \frac{P(M \cap A)}{P(A)}.$$

From the table,

$$P(M \cap A) = \frac{39}{561}, \qquad P(A) = \frac{230}{561}.$$

Therefore,

$$\begin{aligned} P(M \mid A) &= \frac{\frac{39}{561}}{\frac{230}{561}} \\ &= \frac{39}{230} \\ &= \boxed{\frac{39}{230}}. \end{aligned}$$

Be careful! Do not divide by 113. The condition is "40–59," so we divide by the total number of attendees aged 40–59, which is 230. Dividing by 113 would instead give $P(40\text{–}59 \mid \text{Meta})$.

(d) Their company is Apple given that they are not 20–39?

Conditioning on "not 20–39" removes the 20–39 column. The new sample space consists of the 40–59 and 60+ columns, so its size is $230 + 170 = 400$.

	20–39	40–59	60+	Total
Apple	41	53	27	121
Google	24	67	38	129
Meta	34	39	40	113
Other	62	71	65	198
Total	161	230	170	561

$$P(\text{Apple} \mid \text{not 20–39}) = \frac{P(\text{Apple} \cap \text{not 20–39})}{P(\text{not 20–39})}$$
$$= \frac{53+27}{230+170} = \frac{80}{400} = \boxed{0.20}.$$

If you see "if" or "given that" then you are asked to calculate the **conditional probability**!